Yale JACKSON SCHOOL OF GLOBAL AFFAIRS

Final Part I: Export Strategy Due Friday, March 6, 2026

GLBL 5010 Economics for Global Affairs

Submission guidelines: Please submit a zipfile with your code and the final policy memo on *Canvas → Assignments → Final Part I Export Strategy*. Do not submit on Gradescope.

Main Assignment: Write a policy memo for a country of your choice.

The Ministry of Commerce has received a sudden increase in its annual budget. The budget item allows for the development of a new export strategy aimed at improving the country's productivity levels. You have been asked to provide some analysis of the country's current export patterns, and the trade costs it faces. Use your analysis to write a policy memo to recommend how the funds should be targeted to increase the country's exports and improve productivity.

Required Analysis

To complete your memo, you will need to perform several analysis using the following tools introduced in class: (1) Development Accounting, (2) Analyzing Trade Flow Statistics, (3) Comparative Advantage, and (4) Gravity Models. You will need to perform your own analysis of economic data rather than relying on preexisting analyses. You may use previous codes from class or assignments. Each analysis is associated with a section of the policy report. More details of each section are provided below.

Format

The memo consists of several parts:

1. Title
2. Executive Summary
3. Background: Economic Performance

4a. Export and Comparative Advantage Assessment
4b. Export Costs and Strategic Partnership
4c. Potential Impact of Tariffs

5. Recommendations

The policy memo should be a maximum of 3 pages long (excluding graphs) with font size 11, font style Calibri, and 1.5" spacing.

Part 2: Executive Summary

This section should consist of a combination of key findings from your analysis and recommendations. Limit to one to two paragraphs max. If preferred, bullet points are acceptable.

Part 3: Background: Economic Performance
Requirement: One graph (or two graphs side-by-side max) + one paragraph max.

As background information, provide an assessment of the country's recent economic performance. Conduct a growth accounting analysis showing the county's overall growth trend from 1980 to 2019 (use the production function with human capital seen in Assignment 2 question 2). Write a paragraph based on your analysis on how the country has performed and provide a graph of your choice showing how the trajectory of productivity has evolved over time.

Note: You can use your judgement to choose a graph (or two graphs side-by-side) that best highlights key trends that you discussed in the paragraph. It may be one statistic of choice or a combination of statistics. In some cases, it may be helpful to benchmark the country against a comparator country (or set of countries) or highlight different periods in the timeline, for example, periods of different government. Keep in mind also data availability when making your choice. If 1980 or 2019 is not available, use the closest years.

Part 4a: Export and Comparative Advantage Assessment
Requirement: One graph (or two graphs side-by-side max) + one/two paragraphs max.

This section should evaluate the country's export performance. Use the data and tools from class to investigate the following questions: What is the country's export pattern in 2000 versus 2022 (or year closest available)? In which sectors do the country have a strong comparative advantage? In which sectors are its comparative advantages improving or declining? Provide one graph of your choice that best highlights one or multiple key findings from your paragraph(s).

Note: It is fine to focus this discussion on physical goods, instead of services, since it is easier to obtain detailed trade data for physical goods.

Part 4b: Imagining Global Trade without the WTO
Requirement: One table/graph + one paragraph max.

The ministry is interested in your thoughts on the decline of the international rules-based trade system, and specifically, a potential collapse of the World Trade Organization, the agency that has been a driving force towards global trade for three decades. Conduct a gravity model that allows you to estimate the effect of the WTO on the country's exports. Use the latest data from 2022 (or latest available) and estimate a gravity model controlling for: log distance, log market size of destination country, shared border, common language, common legal system, RTA, and WTO indicator of the destination partner. Feel free to add other variables that you think are beneficial to include.

Note: To make sure that you use all countries in your sample analysis, define the log value of exports as log (1+value). Zero values will then not be dropped.

Provide the table of results or a graph of key coefficients from your regression model. Provide a prediction of how much exports would change if the WTO collapsed and went away for all countries. Explain your results in one paragraph.

Part 4c: Barriers to Trade
Requirement: One/two paragraphs max.

Suppose the scenario imagined in 4b happens. Based on your model, what kinds of policies might enable your country to increase exports in a world without the WTO. You do not need to re-estimate the gravity equation. Instead, think about the other covariates in your gravity regression and explain whether there are certain economic policies that can change those covariate levels or their slope estimate to boost exports. Your paragraph(s) should explain what barriers to exports a country should look out for and insight into potential policies to pursue in the future.

Part 5: Summary Discussion

In this section, you should summarize the main findings from the analysis that you did above in one or two paragraphs. Does your analysis lead to any overall policy recommendations on how your chosen country should target funds to increase the country's exports. Your recommendations should logically follow from the claims made in the previous sections.

Optional follow-up questions to consider: Why do you expect your proposed policy to successfully foster exports? What factors would be required for your policy to be successful? What are the potential risks with this policy? How should your policy be implemented to minimize these risks and maximize the chance of success?

Some writing tips:

A good reference book on how to write policy memo is Chrisinger (2022, Chapter 6), "How to Structure Policy Memos and Briefs for Maximum Impact" (available e-book from Yale library). Keep in mind that the sections of this memo differ from the sections discussed in the chapter. Nevertheless, the chapter provides some good advice on writing a memo. Chapters 7 to 9 also provide good advice on writing clearly and persuasively. All chapters are available on canvas.

We do not have a separate "Key Findings" section as discussed in Chapter 6. However, be sure to follow Chrisinger's recommended approach and write each paragraph of the key findings incorporated in Part 3 and Part 4, that is, begin with a doubtable claim (or an argument/ hypothesis), then follow that claim with evidence in the form of data and logical reasoning.

As Chrisinger recommends, write your recommendations using clear, concise, active voice sentences in which you clearly state what actions your chosen country should take (eg, "To increase iron ore exports, Mauritania should...").

Datasets for Analysis

The following are the recommended data sets to be used for analysis. The datasets can be found in the midterm folder on the github class page.

1) final-pwt.dta: This dataset has country-level GDP, population, education, and capital stock from 1950-2019. The source is the Penn World Tables. If you wish to augment these data with additional variables, consider using the World Development Indicators (WDI), which has many

additional country-level variables, such as infrastructure spending, health outcomes and so forth.

Some notes for using this dataset:

- The dataset includes data for 183 countries. It will be helpful to keep only the countries you need for your analysis after loading the file.

2) final-industryflows.dta: I have provided a file that contains countries' exports by industry-year, from 1995-2022. This dataset starts from the data released by UN Comtrade at the product level and aggregates the data to the two-digit HS "industry" level. You only need to work with the industry-level data that I have provided. The list of HS chapters can be found here.

Some notes for using this dataset:

- An observation is an exporter-industry-year (iso_o-hs2-year). The *value* variable indicates the total exports of the country for the industry. The *value_wd* variable indicates total exports of the world for the industry.
- It will be helpful to keep only the country of your analysis after loading the file.

3) final-gravity.dta: This dataset provides bilateral trade between countries, aggregated across all industries, for 1948-2020. So, unlike the "Industry Trade Flows" file, a unit of observation is at the origin-destination-year level. This file contains bilateral gravity variables that may be useful for your analysis in Part 4b and 4c.

Some notes for using this dataset:

- The dataset includes data for many countries. It will be helpful to keep only the country you are analyzing after loading the file. For example, if you want to examine data for the USA, after loading the file, type: `keep if inlist(iso_o,"USA")`. You will now observe exports from the USA to 201 destinations.

- You can find detailed information on the gravity variables here.
    - There is a variable called scaled_sci_2021. This variable captures the intensity of social connectedness between two countries based on anonymized Facebook friendship networks.[1] This metric has been rescaled to range from 1 to 10^9 (for a reason I cannot figure out). Such scaling has no natural meaning, so if you use this variable, I would simply take logs when using it in a regression and interpret it in terms of percent changes.

    - The dataset includes "entry_tp_o" and "entry_tp_d" which are taken from the World Bank Doing Business report and provide an estimate of how many days it takes to start a business in a country; this is a proxy for the business climate in the origin and destination countries. The World Bank does not collect these data for every country, so it may be missing for your country.

    - Some variables are an index between 0 and 1. You can multiply this by 100 to help with the interpretation.
        - "Comrelig" is a religious proximity index bounded between 0 and 1. It adds the product of shares of Catholics, Protestants and Muslims in the exporting and importing countries. "comrelig" = 1 means the country pair has a religion which

(1) comprises a vast majority of the population, and (2) is the same in both countries.

- The variable "rta" is a dummy variable where 1 means that the origin and destination country are engaged in a regional trade agreement of any type within the given year.

- "diplo_disagreement" is the UN Diplomatic Disagreement Score. A higher score means the two countries have a higher disagreement with each other (calculated using UN votes). You can interpret the score as is or standardize the score (subtract the mean, and then divide by standard deviation). The slope interpretation then becomes: "a one standard deviation increase in disagreement score from the mean is associated with ….".

- This file has a trade flow variable that comes from the IMF. It will therefore not match the totals from the final-industry.dta file. You <u>do not</u> need to merge the trade flows from the final-industry.dta file into this file, nor do you need to reconcile the trade flows between the two files.